

Low-Energy Conformer Generation using the Adjoint Schrödinger Bridge Sampler

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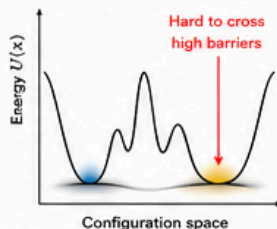
1 Problem: Conformer Generation

Given a molecular system, the goal is to generate conformations distributed according to the Boltzmann distribution

$$p(x) = \frac{1}{Z} e^{-\beta U(x)}$$

where:

- $x \in \mathbb{R}^{3N}$ — atomic coordinates
- $U(x)$ — potential energy
- Z — partition function
- $\beta = 1/(k_B T)$

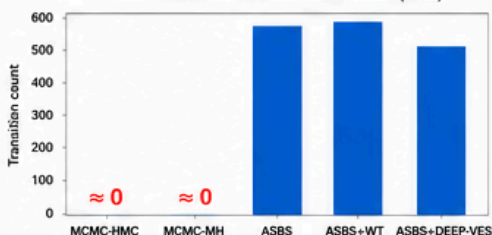


Why is it hard?

- High-dimensional rugged energy landscapes
- Multiple metastable basins separated by high barriers
- Traditional MCMC methods mix poorly
- Low-energy states are rarely visited

We need efficient, general, and amortized samplers capable of globally exploring the landscape.

Transitions Between Metastable Basins (Ala2)



ASBS-based methods produce hundreds of transitions, while traditional MCMC methods almost never leave the initial basin.

2 ASBS: Adjoint Schrödinger Bridge Sampling

ASBS learns stochastic dynamics that transports samples from a simple prior p_0 to the target Boltzmann distribution $p_1(x) \propto e^{-\beta U(x)}$, by solving an entropy-regularized Schrödinger bridge problem.

Optimization problem

Find the controlled diffusion

$$dq_t = f_0(q_t, t)dt + g(t)u_t(q_t, t)dt + g(t)dW_t$$

that minimizes

$$\mathcal{J}(u) = KL(q_T \parallel p_1) + \lambda \mathbb{E}_{q_0} \left[\int_0^T \|u_t(q_t, t)\|^2 dt \right]$$

subject to Schrödinger bridge constraints $q_0 = p_0$, $q_T \approx p_1$.

ASBS alternates between two optimization objectives:

1. Adjoint Matching (AM) — update the control u

$$u^{(k)} = \operatorname{argmin}_u \mathbb{E}_{q^{(k-1)}} \left[\|u_t + \sigma_t (\nabla E(x_t, t) + h^{(k-1)}(x_t, t))\|^2 \right]$$

where $E(x, t) = \beta U(x)$ and σ_t is the diffusion scale.

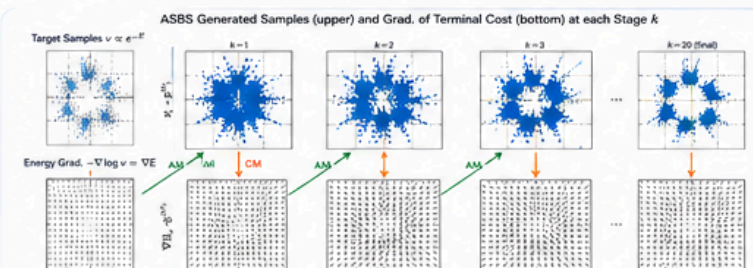
2. Corrector Matching (CM) — update the corrector h

$$h^{(k)} = \operatorname{argmin}_h \mathbb{E}_{(x_0, x_1) \sim q \circ T} \left[\|h(x_1, T) - \nabla_{x_1} \log p^{\text{base}}(x_1 | x_0; T)\|^2 \right]$$

where $p^{\text{base}}(x_1 | x_0; T)$ is the conditional density of the base diffusion.

Total training loss: $\mathcal{L} = \mathcal{L}_{AM} + \gamma \mathcal{L}_{CM}$

Inference: After training, conformers are generated by simulating the learned controlled SDE from $t = 0$ to T to obtain conformers approximately distributed as p_1 .



4 Experiments: Ala2 (Alanine Dipeptide)

Ala2 is a standard benchmark for conformational sampling.

Key properties:

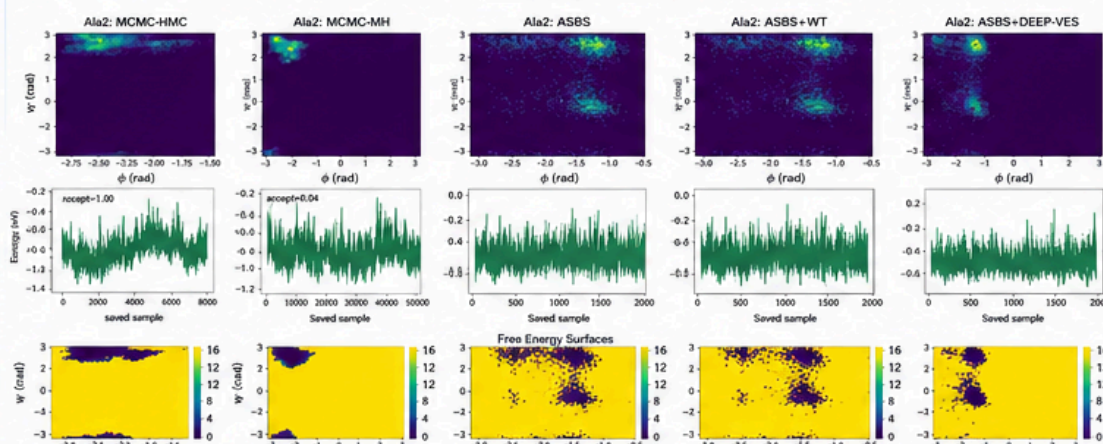
- 22 atoms (66D Cartesian space)
- Two metastable basins:
 - α -helix ($\phi \approx -60^\circ$, $\psi \approx -40^\circ$)
 - β -sheet ($\phi \approx -120^\circ$, $\psi \approx 120^\circ$)
- Strong barriers between basins

Methods Compared

- MCMC-MH (Metropolis-Hastings)
- MCMC-HMC (Hamiltonian Monte Carlo)
- ASBS (Adjoint Schrödinger Bridge Sampler)
- ASBS+WT (Well-Tempered ASBS)
- ASBS+DEEP-VES (Neural Variational Enhanced Sampling)

5 Results

Ala2: MH vs HMC vs ASBS Variants



6 Conclusions

- ✗ Traditional MCMC methods (MH, HMC) mix poorly and fail to explore the full conformational landscape.
- ✓ ASBS discovers multiple metastable basins and captures transitions between them.
- ✓ WT-ASBS improves transition counts using collective-variable biasing.
- ✓ DEEP-VES achieves comparable or better exploration with fewer hyperparameters.
- ➡ Overall, ASBS-based methods significantly outperform traditional MCMC in both efficiency and exploration.

7 References

- ASBS: Adjoint Schrödinger Bridge Sampler
<https://arxiv.org/pdf/2506.22565>
- WT-ASBS: Well-Tempered ASBS
<https://arxiv.org/pdf/2510.11923>
- DEEP-VES: Neural Networks: Based Variationally Enhanced Sampling
<https://arxiv.org/pdf/1904.01305>

3 Biasing for Better Mode Exploration

ASBS can still under-sample very low-energy modes. We introduce biasing strategies using collective variables (CVs).

What are Collective Variables (CVs)?

CVs are low-dimensional descriptors of molecular geometry (e.g., torsion angles, distances) that reduce dimensionality:

$$\mathbb{R}^{3N} \rightarrow \mathbb{R}^k, \quad k \ll 3N$$

For Ala2 we use torsion angles ϕ and ψ .

1. Well-Tempered ASBS (WT-ASBS)

We add a Gaussian bias in the CV space:

$$V_{\text{bias}}(s, t) = -\frac{1}{\gamma} \log \left(\int_0^t e^{-V(s, \tau)/\gamma} d\tau + e \right)$$

where s are collective variables, γ is the bias factor, and $V(s, t)$ is the accumulated (Gaussian hills).

2. DEEP-VES

Instead of depositing hills, we learn a neural bias $V_\theta(s)$ online using DEEP-VES.

- Bias is represented by a neural network $V_\theta(s)$.
- Trained to approximate the bias potential that maximizes exploration.
- Requires far fewer hyperparameters than WT-ASBS.
- Learns smooth bias with good generalization.

We integrate the learned bias into ASBS to obtain **ASBS+DEEP-VES**.

Pros / Cons WT-ASBS

Pros

- Improves transitions between basins
- Widely used in enhanced sampling

Cons

- Many hyperparameters
- Requires hill scheduling
- Does not fully solve amortized sampling

Pros / Cons DEEP-VES

Pros

- Learnable smooth bias
- Stable training
- Better scalability
- Fewer hyperparameters

Cons

- Requires additional neural network
- Extra training stage